

MATH 209 - Calculus 3

Review Formulas

In the table below, $u = u(x)$, $v = v(x)$, $f = f(x)$ and $g = g(x)$ represent four differentiable functions of x , $n \in \mathcal{R}$ and $a > 0$.

$$\begin{array}{llll} y = u \pm v & \Rightarrow & y' = u' \pm v' \\ y = uv & \Rightarrow & y' = u'v + uv' \\ y = \frac{u}{v} & \Rightarrow & y' = \frac{u'v - uv'}{v^2} \quad (v \neq 0) \\ y = (f \circ g)(x) & \Rightarrow & y' = f'(g(x)) \cdot g'(x) \end{array}$$
$$\begin{array}{llll} y = x^n & \Rightarrow & y' = nx^{n-1} & y = u^n & \Rightarrow & y' = nu^{n-1}u' \\ y = \sqrt{x} & \Rightarrow & y' = \frac{1}{2\sqrt{x}} & y = \sqrt{u} & \Rightarrow & y' = \frac{u'}{2\sqrt{u}} \\ y = \frac{1}{x} & \Rightarrow & y' = \frac{-1}{x^2} & y = \frac{1}{u} & \Rightarrow & y' = \frac{-u'}{u^2} \end{array}$$

$$\begin{array}{llll} y = \sin(x) & \Rightarrow & y' = \cos(x) & y = \sin(u) & \Rightarrow & y' = \cos(u)u' \\ y = \cos(x) & \Rightarrow & y' = -\sin(x) & y = \cos(u) & \Rightarrow & y' = -\sin(u)u' \\ y = \tan(x) & \Rightarrow & y' = \sec^2(x) & y = \tan(u) & \Rightarrow & y' = \sec^2(u)u' \\ y = \cot(x) & \Rightarrow & y' = -\csc^2(x) & y = \cot(u) & \Rightarrow & y' = -\csc^2(u)u' \\ y = \sec(x) & \Rightarrow & y' = \sec(x)\tan(x) & y = \sec(u) & \Rightarrow & y' = \sec(u)\tan(u)u' \\ y = \csc(x) & \Rightarrow & y' = -\csc(x)\cot(x) & y = \csc(u) & \Rightarrow & y' = -\csc(u)\cot(u)u' \end{array}$$

$$\begin{array}{llll} y = e^x & \Rightarrow & y' = e^x & y = e^u & \Rightarrow & y' = e^u u' \\ y = a^x & \Rightarrow & y' = a^x \ln(a) & y = a^u & \Rightarrow & y' = a^u u' \ln(a) \end{array}$$

$$\begin{array}{llll} y = \ln(x) & \Rightarrow & y' = \frac{1}{x} & y = \ln(u) & \Rightarrow & y' = \frac{u'}{u} \\ y = \log_a(x) & \Rightarrow & y' = \frac{1}{x \ln(a)} & y = \log_a(u) & \Rightarrow & y' = \frac{u'}{u \ln(a)} \end{array}$$

$$\begin{array}{llll} y = \sin^{-1}(x) & \Rightarrow & y' = \frac{1}{\sqrt{1-x^2}} & y = \sin^{-1}(u) & \Rightarrow & y' = \frac{u'}{\sqrt{1-u^2}} \\ y = \cos^{-1}(x) & \Rightarrow & y' = \frac{-1}{\sqrt{1-x^2}} & y = \cos^{-1}(u) & \Rightarrow & y' = \frac{-u'}{\sqrt{1-u^2}} \\ y = \tan^{-1}(x) & \Rightarrow & y' = \frac{1}{1+x^2} & y = \tan^{-1}(u) & \Rightarrow & y' = \frac{u'}{1+u^2} \\ y = \cot^{-1}(x) & \Rightarrow & y' = \frac{-1}{1+x^2} & y = \cot^{-1}(u) & \Rightarrow & y' = \frac{-u'}{1+u^2} \\ y = \sec^{-1}(x) & \Rightarrow & y' = \frac{1}{x\sqrt{x^2-1}} & y = \sec^{-1}(u) & \Rightarrow & y' = \frac{u'}{u\sqrt{u^2-1}} \\ y = \csc^{-1}(x) & \Rightarrow & y' = \frac{-1}{x\sqrt{x^2-1}} & y = \csc^{-1}(u) & \Rightarrow & y' = \frac{-u'}{u\sqrt{u^2-1}} \end{array}$$

In the formulae below, $u = u(x)$ is a differentiable function, $n \neq -1$ and $a > 0$, and c represents an arbitrary constant.

$$\begin{aligned}\int u^n du &= \frac{u^{n+1}}{n+1} + c, & \int \frac{du}{u} &= \ln |u| + c \\ \int e^u du &= e^u + c, & \int a^u du &= \frac{a^u}{\ln a} + c\end{aligned}$$

In the formulae below $a \neq 0$.

$$\begin{aligned}\int \sin(ax) dx &= -\frac{\cos(ax)}{a} + c, & \int \cos(ax) dx &= \frac{\sin(ax)}{a} + c \\ \int \tan(ax) dx &= -\frac{\ln |\cos(ax)|}{a} + c, & \int \cot(ax) dx &= \frac{\ln |\sin(ax)|}{a} + c\end{aligned}$$

$$\begin{aligned}\int \sec x dx &= \ln(\sec x + \tan x) + c \\ \int \csc x dx &= \ln(\csc x - \cot x) + c\end{aligned}$$

Integration by Parts

$$\int u dv = uv - \int v du, \qquad \int_a^b u dv = uv|_a^b - \int_a^b v du$$

Trigonometric Integrals

Depending on the integral we have, we use some of the following identities.

$$\begin{aligned}\sin^2 x + \cos^2 x &= 1, & 1 + \tan^2 x &= \sec^2 x, & 1 + \cot^2 x &= \csc^2 x \\ \sin(2x) &= 2 \sin x \cos x, & \cos(2x) &= 2 \cos^2 x - 1 \\ & & &= 1 - 2 \sin^2 x \\ & & &= \cos^2 x - \sin^2 x\end{aligned}$$

$$\sin^2 x = \frac{1 - \cos(2x)}{2}, \qquad \cos^2 x = \frac{1 + \cos(2x)}{2}$$

$$\begin{aligned}\sin x \cos y &= \frac{1}{2} (\sin(x - y) + \sin(x + y)) \\ \sin x \sin y &= \frac{1}{2} (\cos(x - y) - \cos(x + y)) \\ \cos x \cos y &= \frac{1}{2} (\cos(x - y) + \cos(x + y))\end{aligned}$$

$$\begin{aligned}\int \frac{du}{a^2 + u^2} &= \frac{1}{a} \tan^{-1} \frac{u}{a} + c \\ \int \frac{du}{\sqrt{a^2 - u^2}} &= \sin^{-1} \frac{u}{a} + c \\ \int \frac{du}{u \sqrt{u^2 - a^2}} &= \frac{1}{a} \sec^{-1} \frac{u}{a} + c\end{aligned}$$

Trigonometric Substitutions

If an integral contains one of the following terms, make the suggested substitution:

$$\begin{aligned}\sqrt{a^2 - x^2} &\implies \text{sub in } x = a \sin t \\ \sqrt{x^2 - a^2} &\implies \text{sub in } x = a \sec t \\ \sqrt{x^2 + a^2} &\implies \text{sub in } x = a \tan t\end{aligned}$$

Partial Fractions

This technique is only used for rational functions of the form $f(x) = \frac{p(x)}{q(x)}$.

If $\deg(p(x)) \geq \deg(q(x))$ then we first apply long division.

After we factorizing the denominator $q(x)$, we use a partial fractions decomposition depending on the factors of $q(x)$ only.

$q(x)$ contains	remark	partial fractions decomposition
$(a_1x + b_1) \cdots (a_nx + b_n)$		$\frac{A_1}{a_1x + b_1} + \cdots + \frac{A_n}{a_nx + b_n}$
$(ax + b)^n$		$\frac{A_1}{ax + b} + \frac{A_2}{(ax + b)^2} + \cdots + \frac{A_n}{(ax + b)^n}$
$(a_1x^2 + b_1x + c_1) \cdots (a_nx^2 + b_nx + c_n)$	$\Delta < 0$	$\frac{A_1x + B_1}{a_1x^2 + b_1x + c_1} + \cdots + \frac{A_nx + B_n}{a_nx^2 + b_nx + c_n}$
$(ax^2 + bx + c)^n$	$\Delta < 0$	$\frac{A_1x + B_1}{ax^2 + bx + c} + \frac{A_2x + B_2}{(ax^2 + bx + c)^2} + \cdots + \frac{A_nx + B_n}{(ax^2 + bx + c)^n}$

Improper Integrals

TYPE I

$$\begin{aligned}\int_a^\infty f(x)dx &= \lim_{t \rightarrow \infty} \int_a^t f(x)dx, \text{ if } f(x) \text{ is continuous on } [a, \infty), \\ \int_{-\infty}^a f(x)dx &= \lim_{t \rightarrow -\infty} \int_t^a f(x)dx, \text{ if } f(x) \text{ is continuous on } (-\infty, a], \\ \int_{-\infty}^\infty f(x)dx &= \lim_{t \rightarrow -\infty} \int_t^a f(x)dx + \lim_{t \rightarrow \infty} \int_a^t f(x)dx, \text{ if } f(x) \text{ is continuous on } (-\infty, \infty).\end{aligned}$$

TYPE II

$$\begin{aligned}\int_a^b f(x)dx &= \lim_{t \rightarrow a^+} \int_t^b f(x)dx, \text{ if } f(x) \text{ has a vertical asymptote at } x = a, \\ \int_a^b f(x)dx &= \lim_{t \rightarrow b^-} \int_a^t f(x)dx, \text{ if } f(x) \text{ has a vertical asymptote at } x = b, \\ \int_a^b f(x)dx &= \lim_{t \rightarrow c^-} \int_a^t f(x)dx + \lim_{t \rightarrow c^+} \int_t^b f(x)dx, \text{ if } f(x) \text{ has a vertical asymptote at } x = c \in (a, b).\end{aligned}$$

If we need to check whether an improper integral converges or diverges (without evaluating the integral), we can use the direct or the limit comparison tests.